

REVIEW

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Mapping the evolution of digital human resource management: a systematic review and bibliometric analysis

Md. Nazmus Sakib^{1*}, Muhammad Shariat Ullah² and Md. Mahbubur Rahman^{2*}

Abstract

Despite the recent, yet rapid advancements in the scholarly field of Digital Human Resource Management (DHRM), comprehensive studies mapping the field's evolution and current knowledge breadth remain limited. To address this gap, the present research aims to comprehend the existing theoretical underpinning, interdependencies, models, and evolving concepts through a rigorous review of the existing literature and identify the current state of development and future research directions. Thus, bibliometric information from 532 documents, published between January 1, 1984, and September 6, 2024, was retrieved from the Scopus and Web of Science Databases. These documents were then systematically analyzed using a two-tier method. The bibliometric analysis revealed a robust annual growth rate in DHRM research, with significant contributions from a diverse range of authors and a notable increase in publications post-2015. Authors from developed countries contributed significantly to the field's development. The content analysis conducted on bibliographic-coupling-based clusters suggests that research in this scholarly area highlights the integration of digital technologies and data-driven strategies in HR processes, emphasizes the strategic use of data analytics in decision-making, and shows the adaptation of HR functions through digital interfaces, along with the alignment of HRM with organizational strategies through comprehensive resource management. From the literature synthesis, this research offers an integrated socio-technical definition of DHRM and a three-layer definition framework (operational, relational, and transformational) that explains how digital technologies reshape HR processes and strategic value creation. However, the study is limited by its selection of keywords that might not capture all relevant DHRM literature, potentially impacting the comprehensiveness and depth of the findings. Regardless, this study complements the existing literature on DHRM by systematically reviewing the area and rigorously implementing the bibliometric and content analysis techniques to comprehensively understand the field. The findings are intended to facilitate future researchers to develop a congenial underpinning of the theories and concepts while harmonizing the development of identified less-developed research clusters.

Keywords Digital human resource management, DHRM, Electronic human resource management, Digital workforce management, Digital human resource transformation, Bibliometric analysis, Systematic literature review

Introduction

Human resource management (HRM) is one of the oldest research areas in the broad academic domain of business and organization [1–3]. Several studies argued that the human resource of an organization is a significant source of competitive advantage [4–6]. Thus, increasing research interest in the HRM field over time is evident [7, 8]. Furthermore, due to the enhanced

*Correspondence:

Md. Nazmus Sakib
mnsakib@du.ac.bd
Md. Mahbubur Rahman
mahbubur.muhit@gmail.com

¹ Department of Management, Faculty of Business Studies, University of Dhaka, Dhaka, Bangladesh

² Department of Organization Strategy and Leadership, Faculty of Business Studies, University of Dhaka, Dhaka, Bangladesh

capabilities for monitoring, controlling, and directing, as well as easy access to technologies (e.g., the internet, mobile devices, and data management infrastructures), ambitious and progressive organizations are gradually leaning toward modern management techniques that can maximize the utilization of the human resources [9–11]. These techniques recently evolved into a new scholarly spectrum called digital human resource management (DHRM), leading to a new research direction in HRM studies [12, 13]. DHRM, which emerged as a promising academic field, has the potential to revolutionize traditional human resource management practices and foster a sustainable competitive advantage for future organizations [11, 13].

According to Bissola and Imperatori [11], El-Kot, and Leat [14], DHRM is the application of advanced information systems technologies and data management strategies, including big data, machine learning, and AI, to enhance and automate traditional HRM processes. In other words, DHRM revolutionized traditional HR practices by implementing modern tools [15–17] and strategies [18, 19]. Additionally, DHRM practices have been found to have positive impacts on workplace safety, health, and sustainable performance [20], collective engagement [21], overall development of resources [22] and innovative economy [23]. Despite being part of a large and popular academic domain, DHRM as a research field has accumulated mere scientific acknowledgments [24], which can lead to misinterpretation of the field's importance and underutilization of its potential. In recent literature, how organizations can implement DHRM focusing on social and technological aspects remains severely in the shadows [9, 11]. However, as the competitive landscape changes due to the COVID-19 pandemic and the synchronization process of the world economy [25], the global market for DHRM-related technologies and interest in research regarding this field are most likely to grow exponentially. As of September 6, 2024, only a few scientific documents provide conceptual backgrounds for studying and understanding the area. Besides, as per the quest for this research, seven of these documents offer detailed and robust literature reviews that have tried to define and conceptualize the term DHRM in a practical sense [9, 10, 12, 14, 26–28], but, significant differences among these conceptualizations can be identified. Further, moderate discussions on the relationships between DHRM and employee performance and satisfaction have been found [29, 30]. This research argues that more insights can be gathered from existing scientific works by conducting a rigorous systematic review and applying bibliometric techniques, which can result in the identification of current trends and future research directions, as well as research gaps in the field of

DHRM. To close these gaps and organize the review, we pose the following research question (RQ).

RQ: How do evolving publication patterns, emerging thematic clusters, and the cumulative theoretical-empirical insights they produce collectively shape and advance the field of DHRM?

The researchers of the current study aim to contribute to the field of DHRM by providing future researchers with a strong understanding of the topic and what has been achieved till now. In line with the RQ, the objectives of this research are threefold.

Objective 1: To map publication trends in Digital Human Resource Management (DHRM) since the field's inception.

Objective 2: To synthesize prevailing themes, patterns, and definitions across existing DHRM studies.

Objective 3: To identify research gaps and propose future research avenues that will advance DHRM scholarship and practice.

Through a systematic literature review and bibliometric analysis, this paper proposes a unified definition of DHRM that can lead to the development of a harmonized and cohesive theoretical or conceptual foundation in future studies. Additionally, the acknowledgment of existing breadth and spread of knowledge will enhance HR practices amidst rapid technological advancements and changing global workplace dynamics. Moreover, the systematic review of the existing studies can facilitate the process of identifying research gaps, sub-areas that have already been explored, and methodological tools that can be used for scientifically studying this field. Again, the acknowledgment of the most prominent authors and sources can guide future researchers to explore relevant scientific works. The bibliometric analysis conducted for this research identified the contribution of different countries, which can be used as a metric for measuring regional evolution and implementation of DHRM that can be useful for determining the most beneficial international collaboration scopes. Finally, future research can use the identified themes to determine the current state of development of the field and future research directions. This study can serve as a valuable resource for scholars seeking to expand their understanding of DHRM and contribute to the advancement of the field.

Defining digital human resource management within the context of the HRM literature

As a research field, HRM primarily revolves around managing human capital through manual endeavors [4, 8, 13, 31]. Previous studies in this field manifested the importance of human resources for organizational success, as well as delineated the ways for managing people within the organizations in the most effective manner

considering the impact and influence of different internal and external variables (e.g., inflation, monetary benefits, organization policies, and national policies) [1, 30–32]. Gradually, the necessity of successful human resource management for sustaining competitive advantage has been recognized by both businesses and researchers globally [33]. Thus, research interest in this field increased [34]. However, recently, organizations globally started leaning toward automation and process innovation due to the increased accessibility of modern technologies (e.g., mobile phones, the internet, personal computers, and servers). This transformation also affected traditional HRM practices to a large extent [9, 24]. Besides, the increased need for organizational efficiencies and productivity to sustain against the exacerbating competitive environment influenced contemporary scholars to merge their view of other academic fields like information technology and data management with HRM, resulting in the emergence of the field of DHRM [11–13, 35].

DHRM has been primarily seen in the previous literature as the outcome of technologies dominating management science. Digital, as a term, has been used in several types of management science-related literature repeatedly in recent times [28, 36, 37]. When defining the impacts of digitalization, the authors notably argued regarding the digital transformation or disruption caused by modern technologies. Meijerink, Boons [38], adopting a cross-disciplinary perspective, indicated the influence of other fields (e.g., big data and artificial intelligence) on DHRM. However, Thite [37] revealed the limitations of technologies, suggesting an application frontier covering only operational and relational activities. The author also identified the lack of focus on value-adding transformational activities in current DHRM literature.

According to Barišić, Barišić [10], *DHRM utilizes recent information systems technology and data management and analysis (e.g., big data, machine learning, AI) to make calculated and rapid organizational decisions*. Nevertheless, DHRM has been characterized differently, catering to varying research needs and contexts. The generalized definition connotes that DHRM is a technical process of converting traditional analog human resource-related data into digital or machine-readable information to automate the management process [12, 26–28]. By combining the contextual differences identified in the studies of Vardarlier and Ozsahin [9], Bissola and Imperatori [11], El-Kot, Leat [14], and Al Haziazi, Muthuraman [35], this research defines DHRM as an aggregation of socio-technical management techniques that integrate automated processes, digitalized infrastructures, and disruptive technologies for managing human resources in tailored environments to satisfy the organizational goal of achieving superior efficiency and effectiveness.

Methodology

Literature reviews in scientific productions have contributed significantly by facilitating the identification of research gaps and knowledge boundaries [39, 40]. Further, these reviews allowed researchers to originate hypotheses and theoretical frameworks for studying any particular area. However, due to their nature, literature reviews enfold multiple and a wide variety of objectives, which is crucial for the seminal intended research activities [41]. Thus, when a focused objective is adopted, researchers utilize tailored versions of tools and techniques for conducting literature reviews [42]. Previous studies in the field of HRM used the systematic literature review approach to apply the iterative process or technique of extracting specified kinds of knowledge [43, 44]. Systematic literature reviews allow the researchers to handle different types of literature published in a field, administer in-depth analysis, and present the findings more comprehensively [45]. Therefore, for this research, the systematic literature review technique has been adopted to study the scientific works published in DHRM. This particular approach of research is also adopted by Jiang, Zhao [46] and two researchers of the present study in their previous research [47]. This systematic literature review aims to comprehensively identify the focus areas, key findings, applied methodologies, and types of analysis and respondents of the mentioned literature to fulfill the broad objective of recognizing the existing prominent research clusters and providing new directions for facilitating the field's development. The summary of this systematic literature review methodology is presented in Figure 1.

Designing and the search query

The previous literature reviews on the area of DHRM, namely the works by Strohmeier [28] and Zhang and Chen [19], facilitated the identification of associated and equivalent keywords for the literature search. Further, three academicians were consulted to verify the appropriateness of these keywords. The search of the literature proceeded after the determination of the study scope, straitening the keywords used as the search query, which were listed as “Digital Human Resource Management” OR “Electronic Human Resource Management” OR “HR Technologies” OR “HR Information Technologies” OR “Digital Human Capital” OR “Electronic Human Capital Management” OR “Technology Integration in HR” OR “HRM Digitalization” OR “HR Analytics” OR “Cloud-based HR Systems” OR “Artificial Intelligence in HR” OR “Machine Learning in HR” OR “HR Software Solutions” OR “Human Resource Management Systems” OR “Talent Management Technology” OR “Workforce Analytics” OR “Smart HRM” OR “Employee Engagement Technology.”

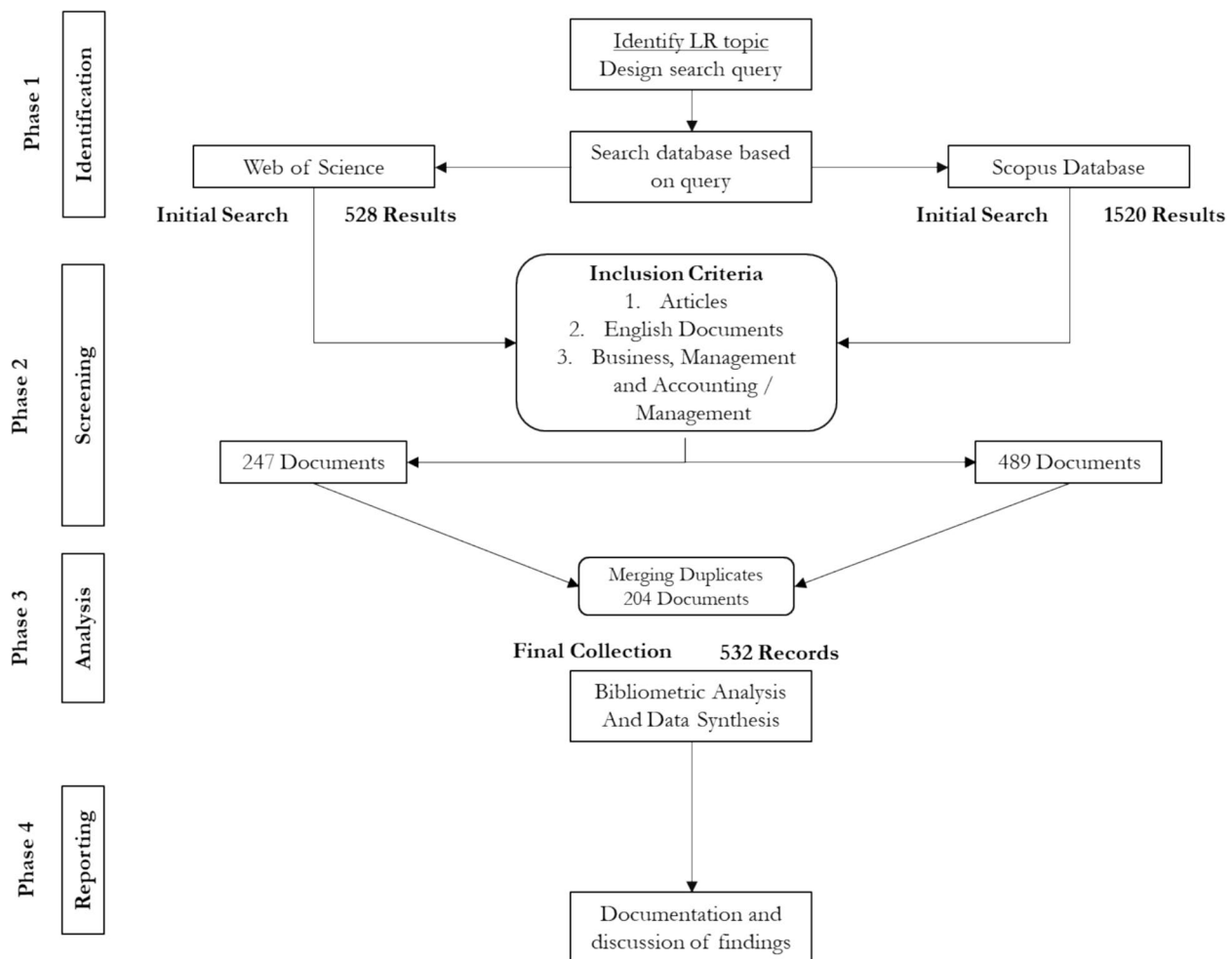


Fig. 1 Summary of systematic literature review methodology

These keywords were used in the title, keywords, and abstract search fields.

Data collection and preliminary results

For conducting the literature search, the Scopus and Web of Science (WOS) databases have been used. The selection of these particular databases is derived from the global acknowledgment as comprehensive and reliable depositories of scientific works [47–49]. Besides, these databases have more peer-reviewed journals in business and management, as well as social science (e.g., PubMed, CORE, and ProQuest) [50]. Table 1 presents the exact search query applied on the databases to extract relevant documents. The search query mentioned in the table initially retrieved 2048 papers published between January 1, 1984, and September 6, 2024. In both databases, the starting point was not defined. Hence, the dataset included the first document that matches the search query.

Refinement of the results

Table 1 also presents the inclusion criteria that were imposed to refine the results. In both cases, refinement was conducted through the default filtering system of the mentioned databases. Previous studies that conducted systematic literature reviews mostly acknowledged peer-reviewed journals as certified knowledge. Despite being an apparent broad research area, it is still in its emergent status, ascribing to its nature of being dependent of technological advancements, which proliferated in recent generations [13, 14, 51]. However, as the scholarly field remains at its emergent status similar to comparable areas [52], a moderate number of articles published in the field was estimated. However, to include only the evidence and insights into current research frontiers, only the journal articles were considered. Other types of documents retrieved from the database were illuminated. The numbers of documents remained after setting each criterion is also presented in the mentioned table. More

Table 1 Search queries and inclusion criteria

Databases and criteria	
<i>Search query</i>	
Scopus	TITLE-ABS-KEY ("Digital Human Resource Management" OR "Electronic Human Resource Management" OR "HR Technologies" OR "HR Information Technologies" OR "Digital Human Capital" OR "Electronic Human Capital Management" OR "Technology Integration in HR" OR "HRM Digitalization" OR "HR Analytics" OR "Cloud-based HR Systems" OR "Artificial Intelligence in HR" OR "AI in HR" OR "Machine Learning in HR" OR "HR Software Solutions" OR "Human Resource Management Systems" OR "Talent Management Technology" OR "Workforce Analytics" OR "Smart HRM" OR "Employee Engagement Technology") AND (LIMIT-TO (SUBJAREA, "BUSI")) AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English"))
WOS	("Digital Human Resource Management" OR "Electronic Human Resource Management" OR "HR Technologies" OR "HR Information Technologies" OR "Digital Human Capital" OR "Electronic Human Capital Management" OR "Technology Integration in HR" OR "HRM Digitalization" OR "HR Analytics" OR "Cloud-based HR Systems" OR "Artificial Intelligence in HR" OR "Machine Learning in HR" OR "HR Software Solutions" OR "Human Resource Management Systems" OR "Talent Management Technology" OR "Workforce Analytics" OR "Smart HRM" OR "Employee Engagement Technology")
<i>Search and refinement</i>	
Scopus	Total documents: 1520 Limited to the field: Business, Management and Accounting (733 documents) Limited to document type: Article (499 articles) Limited to Language: English (489 articles)
WOS	Total document: 528 Limited to document type: Article (383 articles) Limited to WOS Categories: Management (216 articles); Business (61 articles); Total: 248 articles Limited to Language: English (247 articles)
Remove duplicates	204 duplicates removed
Final dataset	532

specifically, only journal articles in English are included in the final dataset, and the conference papers, conference reviews, book chapters, review articles, and non-English documents were removed from the final database. This refinement reduced the total number of documents to 489 and 247, in Scopus and WOS, respectively. The bibliographic details (i.e., the authors' names, abstracts, keywords, affiliations, citations, and references) were stored in CSV format. Finally, the researchers employed programming language R to merge the dataset and remove duplicates. In total, 204 duplicates were removed. Hence, the final dataset consisted 532 documents.

Bibliometric analysis and systematic literature review

Bibliometric analysis has been conducted on the final dataset to capture the breadth of the research field. This analysis technique is a quantitative tool to measure aspects primarily related to dependency, similarity, and differences among written documents. Previous studies powered by bibliometric analysis have found both current research trends and future research directions, as well as fields' development trends and contributions of different entities and sources [24, 34, 48]. However, unlike bibliometric analysis, the systematic or structured

literature review can provide in-depth knowledge, which is crucial to understand the field's current state comprehensively. As Linnenluecke, Marrone [53] suggested, bibliometric analysis and literature review techniques can complement each other to facilitate the extraction of insights related to the depth and breadth of any particular research field. Therefore, this research adopted both of these techniques. The retrieved papers were analyzed using bibliometric statistical tools and content analysis techniques. The R software's bibliometrix package and NVivo 14 were used for bibliometric analysis and literature review, and VOSviewer was used for the network analysis.

Following the bibliometric mapping, this research specifically incorporated clustering techniques during the bibliometric analysis phase to facilitate the systematic literature review process. For clustering, keyword co-occurrence and bibliographic coupling analyses were conducted using VOSviewer. This enabled the identification of thematic clusters, which grouped articles that share strong similarities in topics, methods, or focus areas. Each cluster represented a distinct research stream or thematic area within the broader DHRM field. Following the clustering, a targeted qualitative review was

performed for each cluster: abstracts, keywords, and, where necessary, full texts of representative papers within each cluster were systematically analyzed and recorded using NVivo 14. This two-step process (i.e., quantitative clustering followed by in-depth qualitative synthesis) provided both a high-level structural overview of the field and a detailed understanding of the research themes, methodological approaches, and knowledge gaps.

Analysis and findings

Summary statistics

This section summarizes the findings of the bibliometric analysis. As presented in Table 2, the analysis revealed the contribution of 259 sources, including both domestic and international publishers, and recognized the contributions of 1188 individuals who authored 532 documents between January 1, 1984 and September 6, 2024. The average annual growth rate of documents published in the field was 11.25%, and the average citations per document were 24.23. The average age of the documents was only 6.35 years, indicating a recent proliferation of the field. The documents frequently cited other scholarly works, with an average reference of 49.11 previously published documents. Moreover, collaboration among contributors was also observed, with an average of 2.67

individuals per document. However, the international collaboration index is moderate at 14.48%.

Performance and citation trend analysis

The publication of the first document on DHRM dates back to 1984 [54], and the second publication followed 4 years later [55]. The databases included almost 99% documents published in this field from 1995 to 2024, indicating the actual age of the associated scholarly area. As illustrated in figure 2, the annual publication trend has been inconsistent over the past 27 years, with a marked increase in the number of publications observed only after 2015, with a slight drop in 2018 and 2020. However, the number of publications significantly increased in 2020. Applying Pareto's law, more than 80% of scholarly papers on DHRM have been published in the last 15 years (2009-2024), indicating the domain's youthfulness and which justifies the significant gap in comprehending DHRM in broader contexts. Figure 3 presents the analysis of average annual citations, which indicates an asymmetric, with the highest average annual citations observed in 2011, followed by a steady decline in subsequent years.

Contribution by countries

The results showcase a diverse array of nationalities contributing to the field, reflecting the influence of various cultural and contextual factors. Despite its nascent stage, DHRM has garnered global interest, new authors bringing unique perspectives on situational and cultural factors. The top 10 countries in terms of publications, as detailed in Table 3, are led by the USA with 60 documents, followed closely by India with 47 publications. Authors from the USA started publishing way earlier than authors from other countries, as shown in Figure 4. China also made a substantial impact with 35 publications, including a notable increase in contributions from 2020 to 2024. Australia and Spain contributed 33 and 23 publications, respectively. Germany, the United Kingdom, and Malaysia also made significant contributions with 22, 18, and 16 publications, respectively. Italy and the Netherlands each contributed 14 documents to the field. The low international collaboration suggests potential for further global knowledge exchange and collaboration.

Impact of sources and institutes

As argued by Sahabuddin, Sakib [49], impact of a source can be measured by its H-Index. Figure 5 presents the impact of different sources in the academic field of DHRM, measured by the H-index. The International Journal of Human Resource Management accumulated the highest H-index, receiving 19 within the researched

Table 2 Summary statistics

Description	Results
<i>Main information about the data</i>	
Timespan	1984:2024
Sources (Journals, Books, etc.)	259
Documents	532
Annual growth rate %	11.25
Document average age	6.35
Average citations per doc	24.23
References	26128
<i>Document contents</i>	
Keywords plus (ID)	906
Author's keywords (DE)	1597
<i>Authors</i>	
Authors	1188
Authors of single-authored docs	85
<i>Authors collaboration</i>	
Single-authored docs	105
Co-authors per doc	2.67
International co-authorships %	14.1
<i>Document types</i>	
Article	516
Article; early access	16
Total	532

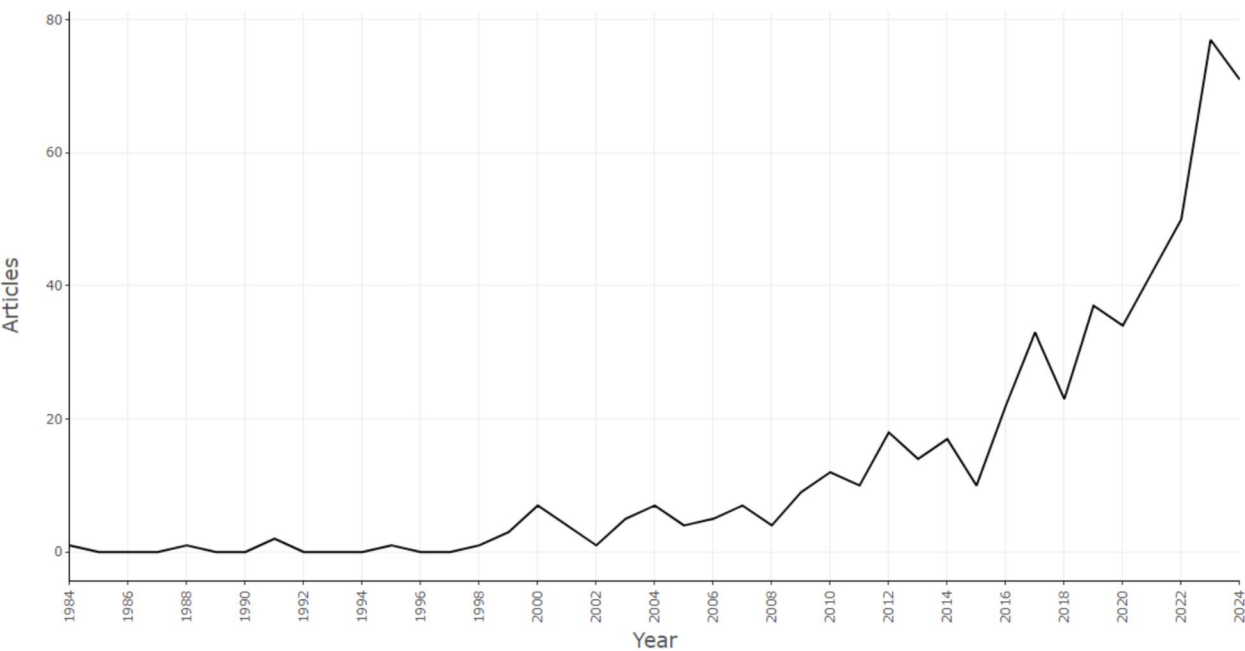


Fig. 2 Trend in annual scientific productions

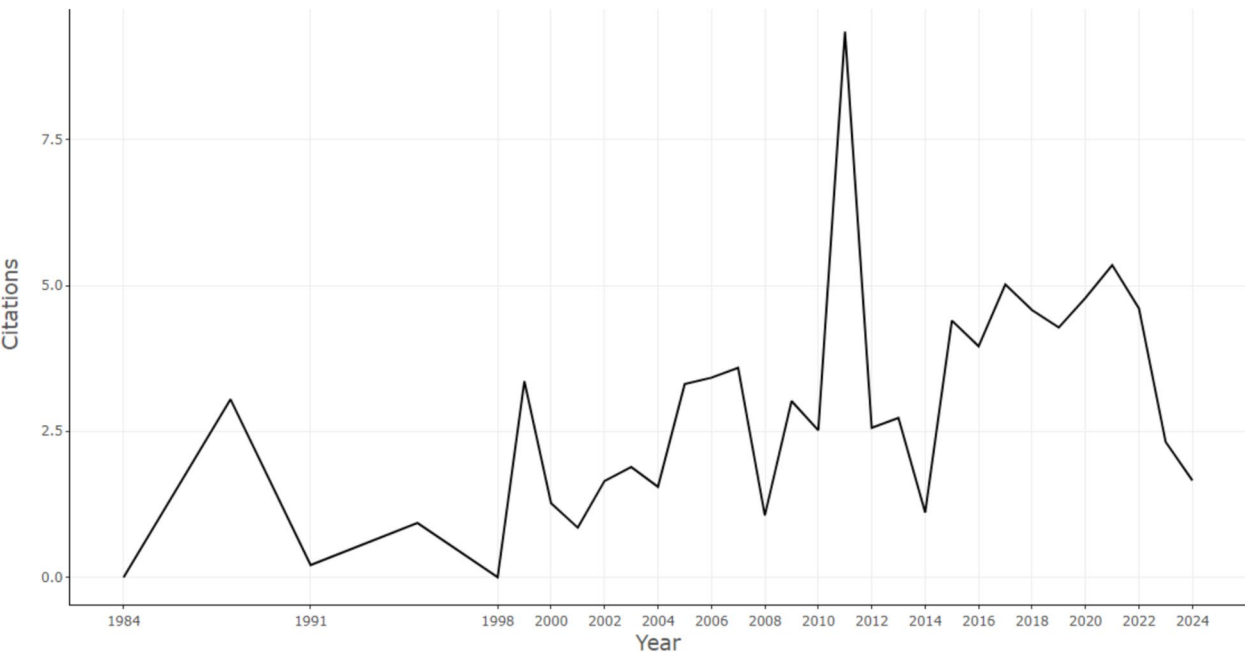


Fig. 3 Annual average citations

period. Human Resource Management followed with an H-index of 16, while Personnel Review secured the third position with an H-index of 11. Human Resource Management Review and the International Journal of

Manpower both received an H-index of 10. Journals such as the Human Resource Management Journal and Journal of Organizational Effectiveness-People and Performance also received substantial H-index scores, further indicating their contribution to the field.

Table 3 Total publication numbers by countries

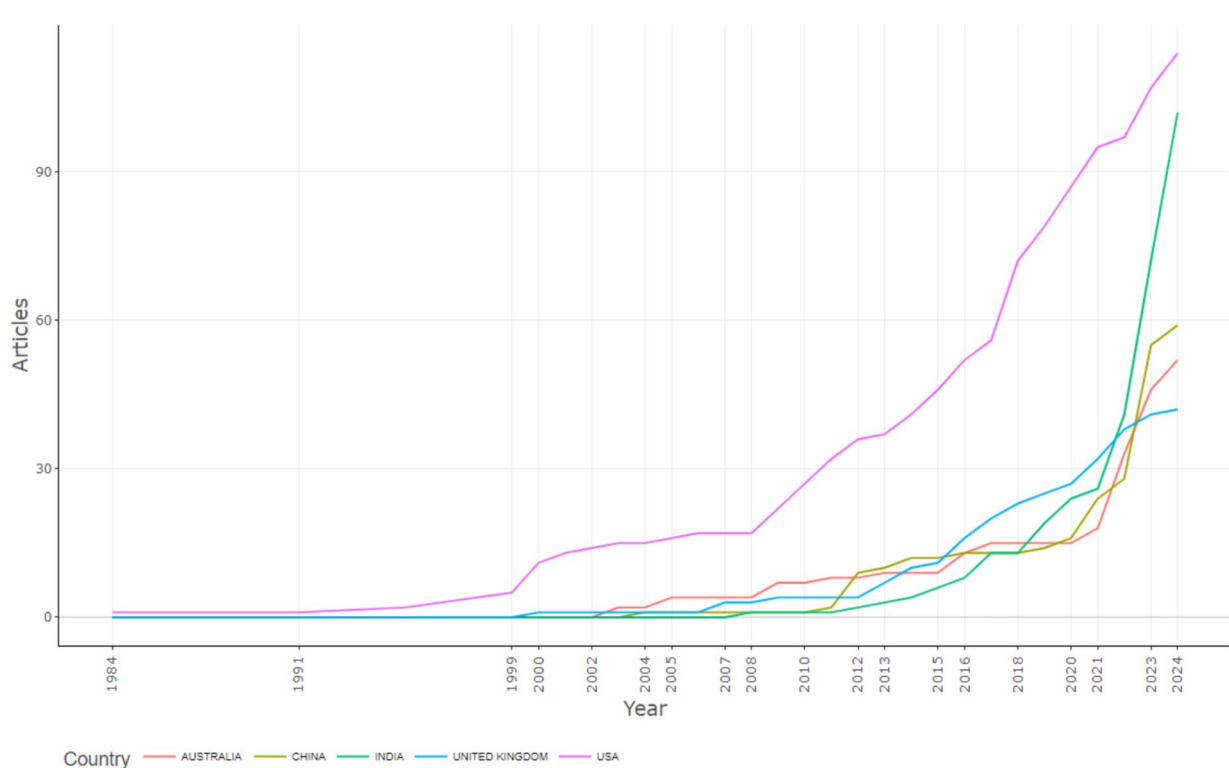
Countries	Number of publications
USA	60
India	47
China	35
Australia	33
Spain	23
Germany	22
United Kingdom	18
Malaysia	16
Italy	14
Netherlands	14

As indicated in Figure 6, Remnin University of China, Saarland University, and the University of Twente emerged as the most relevant affiliations in the field, each with ten publications. Royal Melbourne Institute of Technology (RMIT) and Pennsylvania Commonwealth System of Higher Education (PCSHE) followed, each with nine publications, indicating a significant contribution of these institutes to the field. Other institutes listed in the figure are from various countries with a lower number

of total production, highlighting the scope for other research institutes to make significant contributions to the field.

Impact, relevancy, and contribution of the authors

The bibliometric analysis also yielded valuable insights into the contributions of individual authors to the field of DHRM. The study identified a total of 1188 authors, with Strohmeier S ranking as the highest contributor with nine published documents in the field (Figure 7). Additionally, Bondarouk T published a significant number of documents, indicating the magnitude of the author's work to the development of the field. Ferraris A, Florkowski GW, Hong Y, Marler JH, and Zhou Y each contributed five publications. Figures 8 and 9 present the analysis of total citations (TC) and H-indexing, which further evaluated the impacts of different authors, with Strohmeier S and Bondarouk T receiving the highest scores in H-index. The insights gained from this analysis are of significant academic value, allowing researchers, practitioners, and policymakers to gain a better understanding of the contributions of different authors and their impact on the growth of the DHRM field.

**Fig. 4** Trend in the publication numbers of the countries

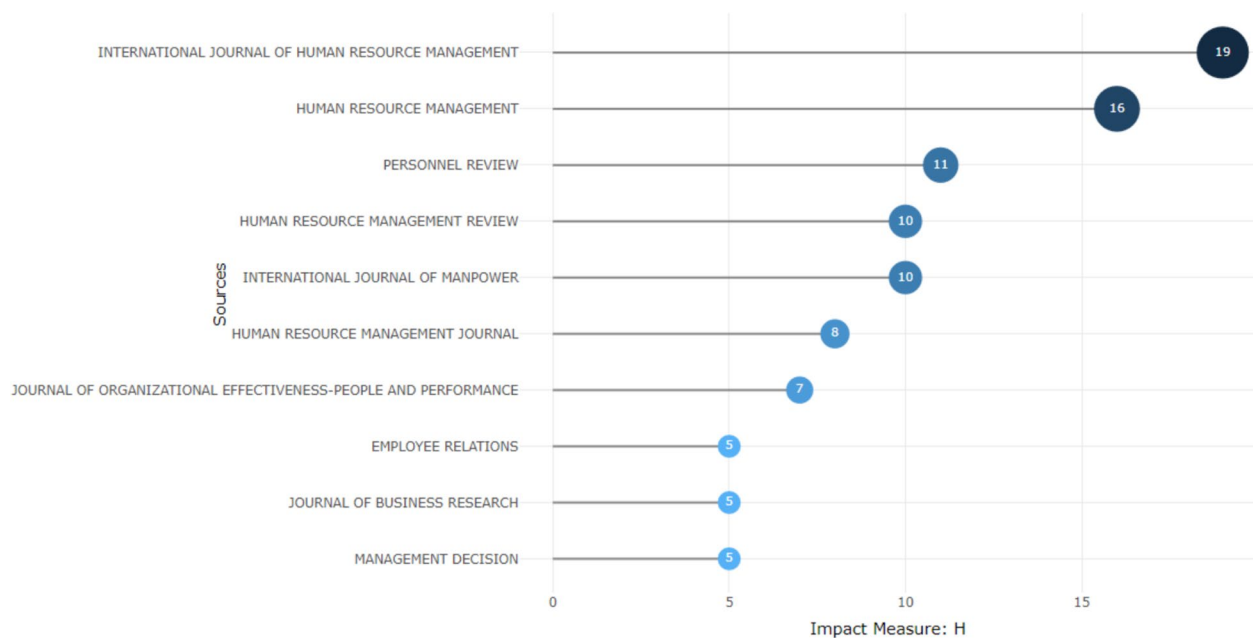


Fig. 5 Local impact of the sources

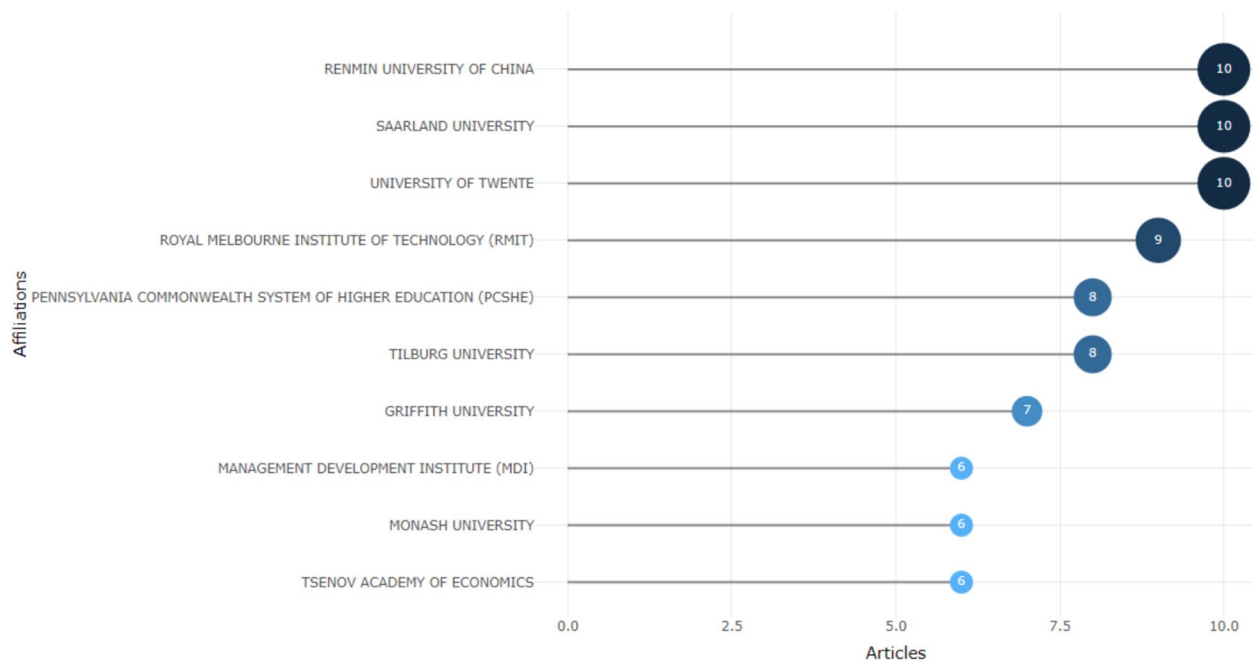


Fig. 6 Most relevant affiliations

Local and global impacts of the documents

Local citation implies a document's impact on the academic area related to the document itself [53]. Figure 10 presents the list of the top ten most locally cited documents in the specified academic field till 06th October 2024. The document authored by Marler and Boudreau

[56] ranks first with 77 local citations, and a document by Angrave, Charlwood [57] follows closely with 68 citations. Strohmeier [44] work also shows significant impact, gathering 60 local citations. These documents illustrate the considerable influence they have within the scholarly field. This further supports the prominence of

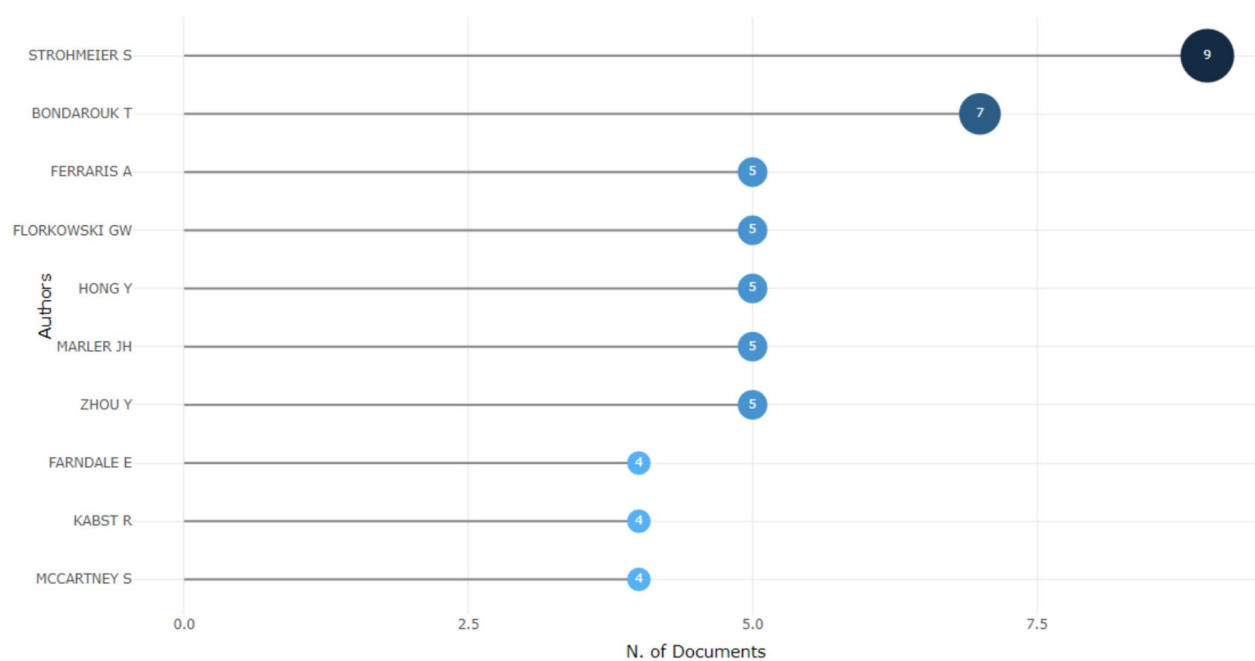


Fig. 7 Publication numbers of the authors

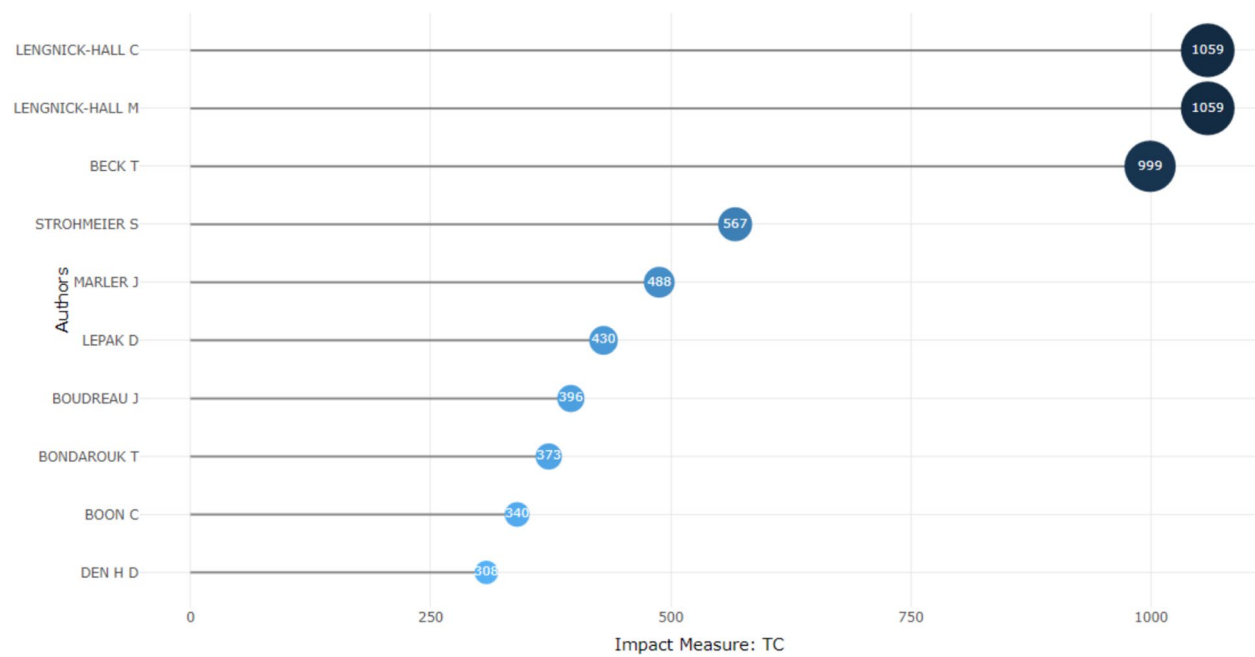


Fig. 8 Most locally cited authors

these authors, affirming their substantial role in shaping the development of the field of DHRM. Rasmussen and Ulrich [58] authored another notably cited document, ranking fourth with 51 local citations, and published in “Organizational Dynamics” in 2015. The visible

contributions of other authors to the development of the field are also evident in the figure, showcasing a diverse range of impactful scholarly work. Global citations of a document indicate the extent of the impact on different related and unrelated academic

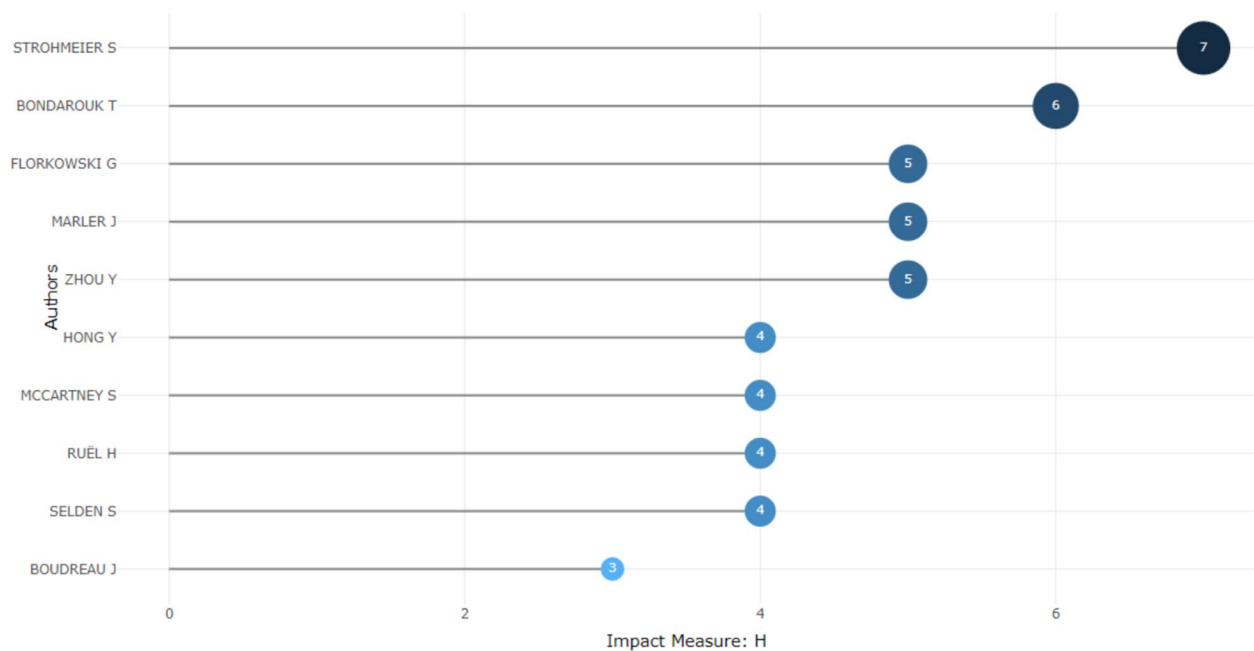


Fig. 9 Impact of the authors

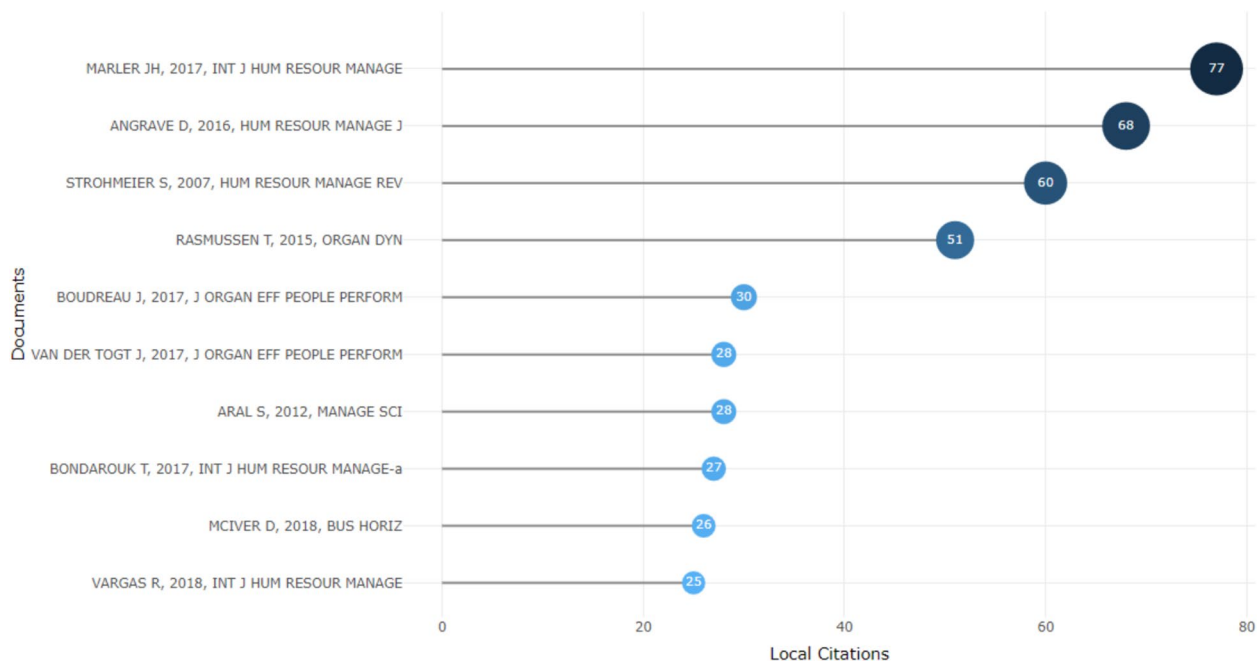


Fig. 10 Most locally cited documents

areas. Figure 11 presents the top 10 most globally cited documents in the field of DHRM during the mentioned period. The document authored by Lengnick-Hall, Beck [59] accumulated the highest number of global citations (999) till October 6, 2024. Strohmeier [44] authored the

second most globally cited document in the field, which garnered 329 global citations till the end of the mentioned research period. The document authored by Marler and Boudreau [56] ranked fourth for accumulating 328 global citations. Most of the significant globally cited

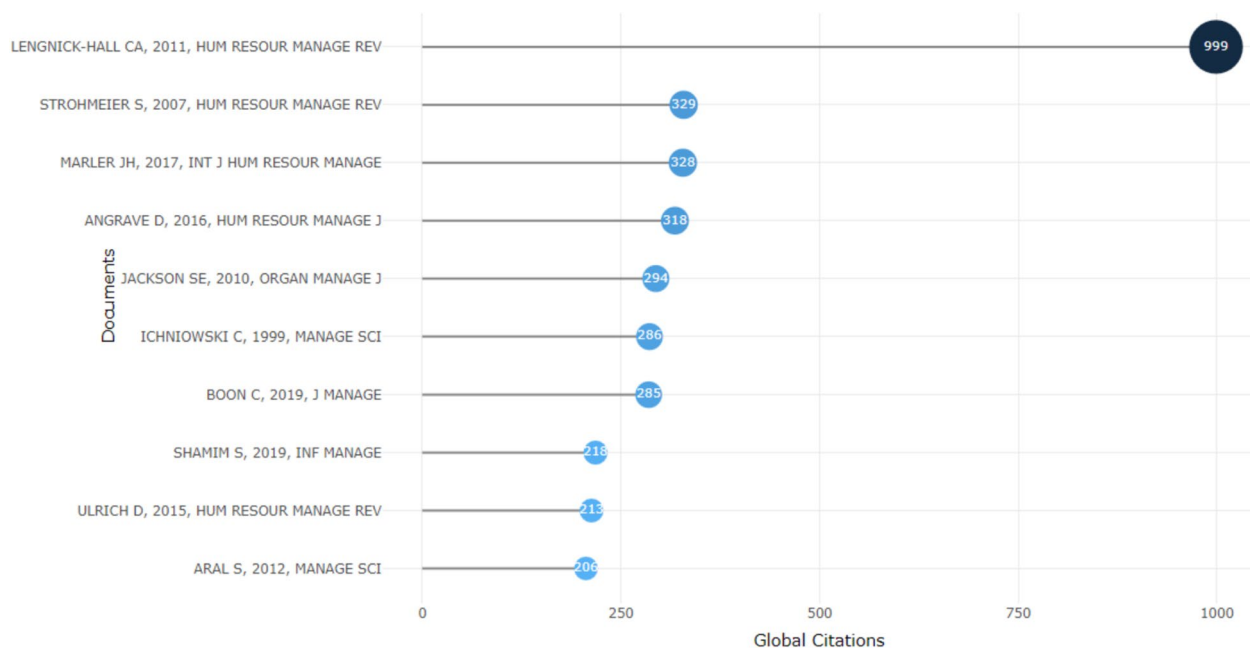


Fig. 11 Most globally cited documents

individuals, including the authors of the top 3 most globally cited documents, are from the USA, Germany, and other developed countries.

Relationship between most cited research papers, authors, and keywords

The three-field analysis conducted for the research offers a detailed view of the connections among authors, keywords, and the most referenced works in the field of DHRM. Figure 12 demonstrates how influential authors are linked to specific keywords and frequently cited documents, highlighting the key areas and foundational texts that shaped or are shaping the field. In the left column (AU), prominent authors such as Strohmeier S and Marler JH are listed, showcasing those whose research significantly impacts the field. The middle column (DE) details the keywords most commonly associated with these authors, such as “e-hrm” and “human resource management.” These keywords indicate the primary themes within their research. The right column (CR_SO) represents the most referenced documents within the field. This illustrates the foundational works that continue to influence ongoing research in DHRM. The high frequency of references to certain works like those by Marler JH and Strohmeier S in top-tier journals underscores their importance and the reliance of the field on these seminal studies. Keywords like “e-hrm” feature prominently, reflecting the field’s focus on the digital transformation of HR practices. Other important keywords include “hr analytics” and “electronic-human resource

management,” pointing to a strong interest in how technological advancements are integrated into HR functions.

Bibliographic links of the academic journals

The bibliometric analysis also identified conceptual clusters in the scholarly field of DHRM. As illustrated in Figure 13, these clusters are noticeably different. The bigger cluster (colored red in the figure) is primarily about systems-level themes within HRM. This cluster also includes research sub-areas: human resource management, human resource management systems, and information management. Another cluster (colored blue in the figure) comprises smaller research sub-areas: human, employee, and resource management. The clusters indicate that the existing scientific works in the field of DHRM focus more on organizational resources and capabilities (e.g., knowledge management infrastructure, information system, and big data). A significant gap in the sub-areas like employee satisfaction, engagement, and retention, as well as outcomes of DHRM, can be identified in the researched documents. Future scholars can contribute to these sub-areas to diversify the research clusters.

Themes in the academic products

The thematic mapping conducted for this research highlighted significant focus areas within the field of DHRM. Figure 14 provides a thematic map that categorizes various themes according to their centrality and development stage within the field. The core themes identified are predominantly focused on “human resource management,”

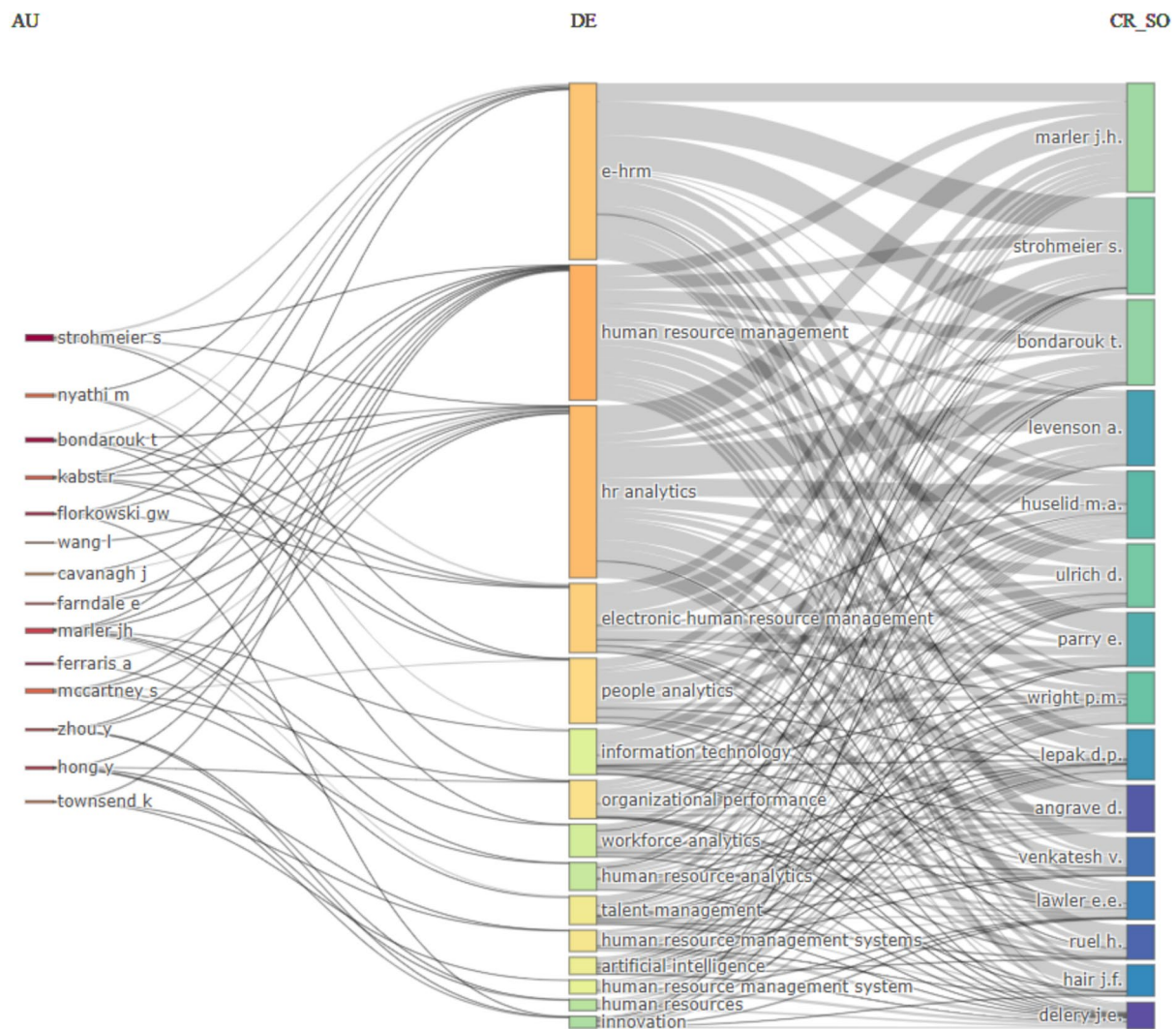


Fig. 12 Three-field analysis in DHRM

“human resource management systems,” and “information management.” Together, these keywords indicate that these areas are central and well-developed within the field. These areas, represented in the motor themes section of the map, suggest that most scholarly attention is given to the integration of information systems within human resources management practices. Additionally, emerging or declining themes identified in the analysis include “managers” and “decision making,” indicating these topics are gaining or losing traction within the academic discourse. These might represent newer areas of interest or previously popular areas that are receiving less focus over time [60]. Niche themes, which might be specific but less developed in terms of centrality in the field, include “resource management,” “human,” and “employee.” These represent specific areas that may be explored further by future research. Furthermore,

the analysis indicates that themes like “competition” and “personnel” are foundational but are not as central as the motor themes. This suggests that they are important, but they do not currently represent the main focus of DHRM research, as the research on these sub-areas are not tightly integrated with previous works.

Finally, to further identify the focus areas of previous scientific works, the researchers of the present study conducted bibliographic association-based clustering in VosViewer. This clustering provides four distinct themes within the scholarly area of DHRM. The themes are highlighted in different colors in Figure 15.

Innovative HRM paradigms (RED)

This research sub-area emerged as the most dominating one within DHRM literature. The documents in this cluster, for example the works of Strohmeier [28], Gupta,

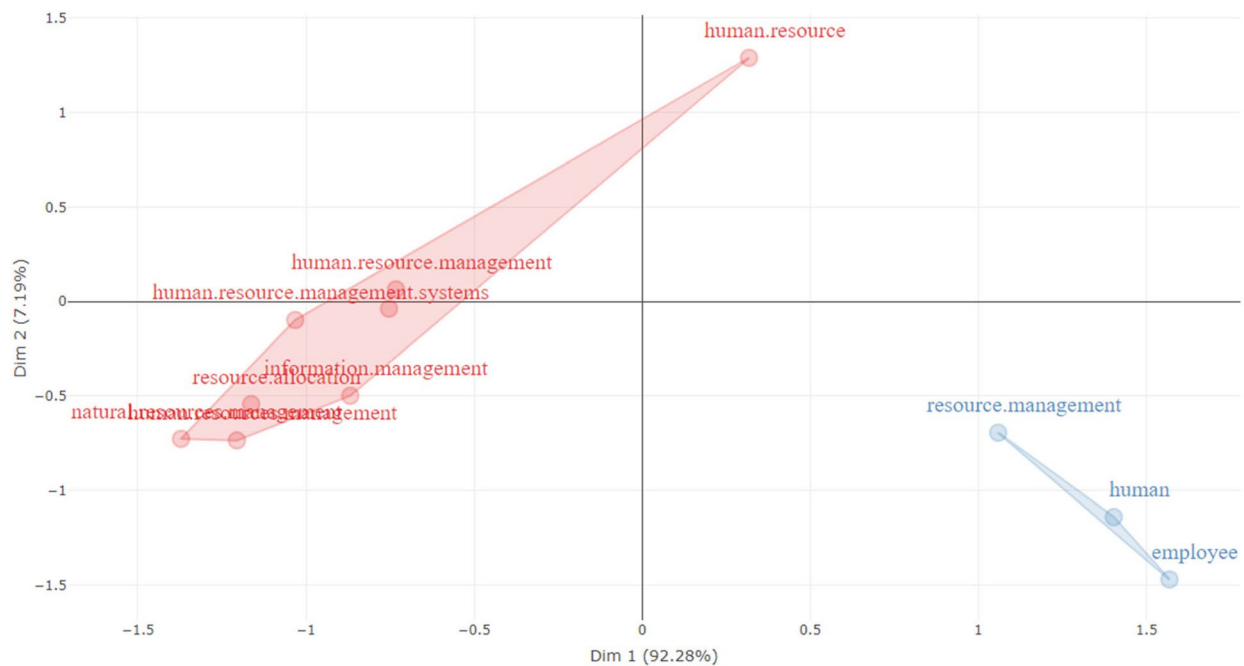


Fig. 13 Cluster analysis in DHRM. Note: conceptual structure map method: multiple correspondence analysis

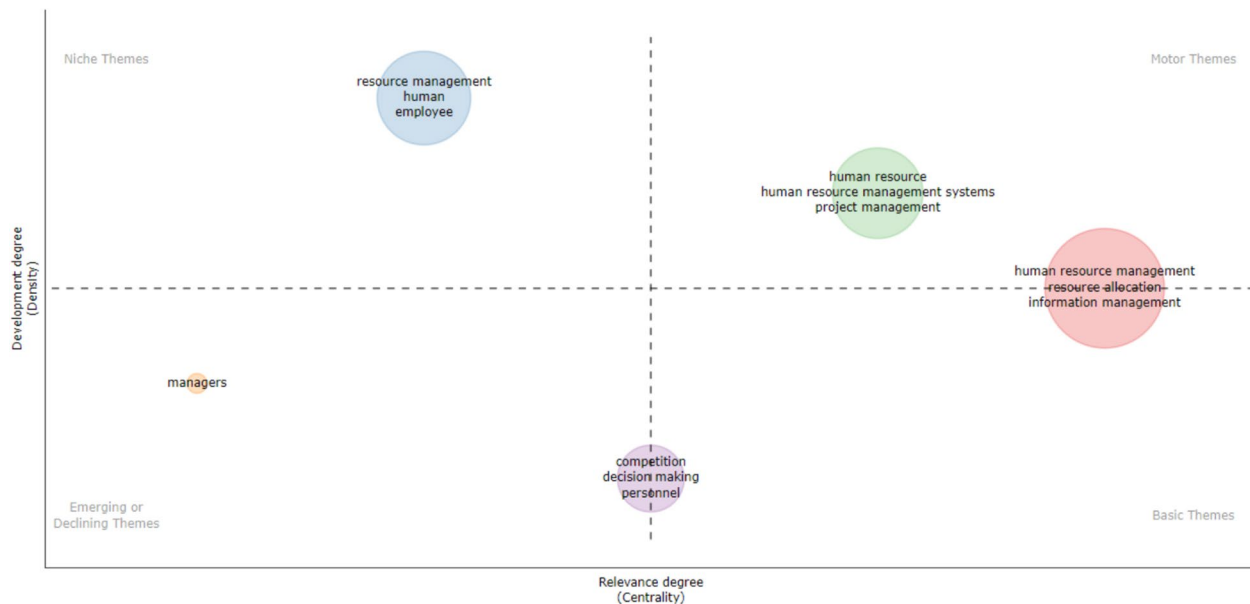


Fig. 14 Thematic map in DHRM

Hassan [61], Böhmer and Schinnenburg [62], Dasari and Devi [63], primarily focused on revolutionizing traditional HR practices through the integration of digital technologies. As depicted in Figure 15, this cluster contains keywords, such as “e-hrm,” “human capital,” “performance,” “innovation,” and “electronic human resource management.” These evidence presented by the associated documents signify a shift from conventional HR

methods to more dynamic, technology-driven strategies [64, 65]. This shift is particularly visible in the application of e-HRM systems. According to Von Glinow and Teagarden [55], Frauenheim [66], Gueutal, Strohmeier [67], and Priyadharshini, Kamalanabhan [68], the mentioned system streamlines HR processes; thus, it enhances overall operational efficiency and improves employee satisfaction. Again, the linkage with “human capital” and

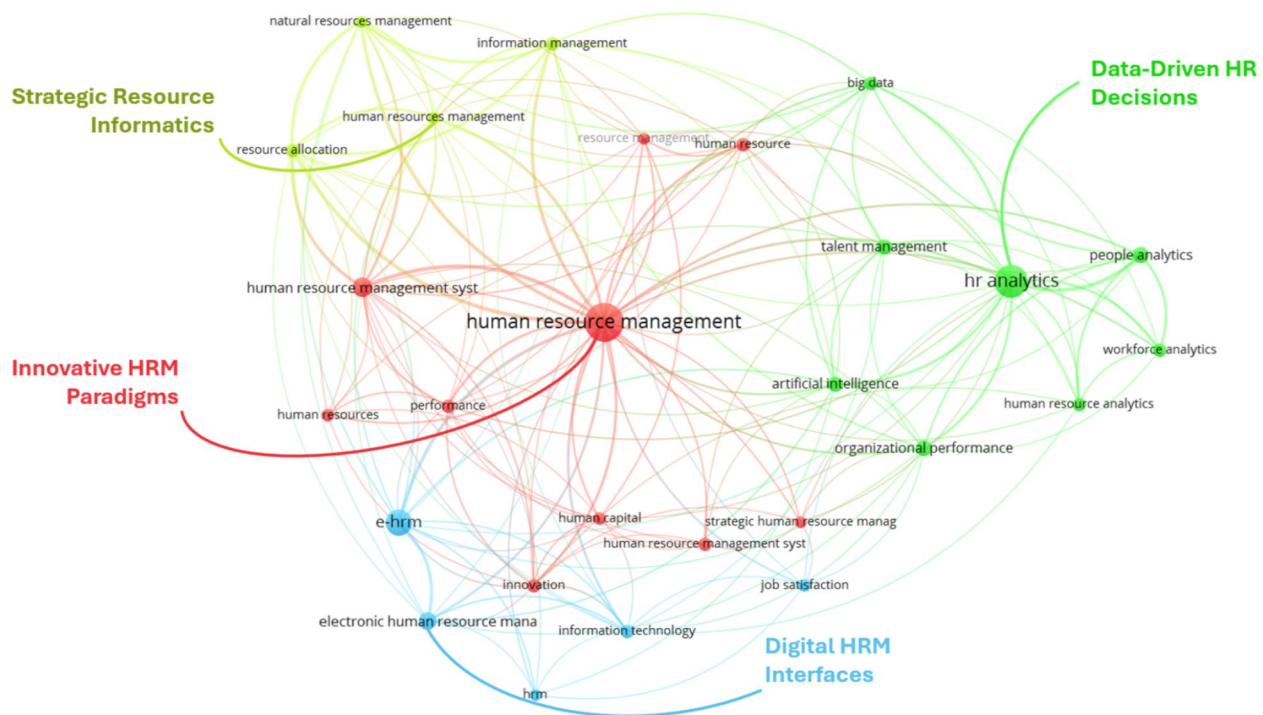


Fig. 15 Bibliographic association-based clusters

“performance” underscores the strategic intent of innovative HR paradigms to optimize talent management and align human resources with organizational goals more effectively [69–71]. Furthermore, the documents within this cluster emphasize on innovation [72], which highlights the creative and forward-thinking approaches in HRM [73], such as the adoption of AI-driven analytics for predictive talent management and enhanced decision-making processes [74–77].

Data-driven HR decisions (green)

Data-driven HR decisions, as argued by Belizón, Majarín [78], Schroeder [79], Gurusinghe, Arachchige [80], is one of the primary reasons for DHRM transformation among modern organizations. Ascribing their fundamental roles, if not the purpose as a whole, DHRM tools, for instance, big data and analytics, can profoundly influence HR practices [81, 82]. As visualized in Figure 15, this cluster associates with keywords such as “HR analytics,” “people analytics,” “workforce analytics,” and “human resource analytics.” The documents encompassing these keywords suggest that strategic decision-making in HR is the systematic use of data derived from various sources, including but not limited to employee performance metrics, recruitment process, and retention rate, to inform and optimize HR strategies [83–85]. For instance, HR analytics tools can predict candidate success rates [86],

identify high-risk turnover groups [87, 88], or pinpoint leadership potential within existing personnel [89]. This analytical approach, from practical point of view, refines talent management, while also closely aligning organization’s strategic objectives in HR policies and practices [79, 80]. Existing studies also presented the application of information and communication technologies in organizational control systems [90]. Besides, the capabilities provided by big data or AI-based predictive mechanisms, can help HR professionals to manage, engage, and motivate employees effectively [91, 92].

Digital HRM interfaces (blue)

Scholarly works in this theme evidence the integration of advanced digital platforms and interfaces in management human resources. The central theme of this cluster is the E-HRM. This platform facilitates automation and optimization of traditional HR functions through digital solutions [93, 94]. According to Blom, Du Plessis [95] and Suri and Lakhanpal [96], this integration enhances accessibility and efficiency, enabling seamless communication and data management across various HR activities. Besides, the benefits of these platforms are already well-addressed. For example, apart from the productivity enhancements, these tools enriches the employee experience. This approach makes HR practices more adaptable

and employee-friendly in contemporary work environments [97].

Strategic resource informatics (neon-green/yellow)

The studies within this cluster indicates the integration of information systems into HRM to optimize resource allocation and enhance organizational capabilities. Though employees are the most critical resource of an organization, strategic HRM focusing only on the people can never sustain [5]. Organizations need to focus on both human and natural or physical assets [98]. This focus aligns HRM with organizational strategies and business objectives. Through resource management plans and DHRM tools, organizations can gain a comprehensive overview of their resource distribution and utilization [99, 100]. In other words, the inclusion of broader resource management system within DHRM, expands the scope of HR's role in managing a broader array of organizational resources beyond just human capital. A recent keyword-co-occurrence analysis of AI-and-IoT research likewise shows how digital technologies are transforming established practices [101]; regardless, the strategic transformation of businesses due to DHRM is rarely to be observed [57, 73, 102].

Discussions and practical implications

Similar to previous bibliometric studies [103, 104] on the field of DHRM, analyses presented in the previous sections indicate an upsurging growth in publication quantity, as well as increasing research interest in the field. This research contributes to the literature on DHRM by adding reviews of recent publications and providing insights regarding current research trends and future research directions. The bibliometric analyses, integrated with the review of DHRM literature (Table 4), enabled the authors to comprehensively identify and present prominent research topics, prolific authors and sources, major research regions and institutes, and the current state of development. These analyses identified noticeable contributions of only a few research institutes and authors. As a field under the umbrella of management science, the impact of the documents published in the field is more pronounced globally. However, considering the relatively low number of documents published in the field since its inception, the interdependency of these documents is also noticeable. Though the first document on DHRM was published in 1984, the field's proliferation has not begun until 2015. More than 80% of the documents were published from 2009 to 2024, signifying the young state of development. Regardless of this proliferating stage, the top three most impactful documents in this field were published between 2007 and 2017. Further, the authors from USA published the highest number of documents.

Despite authors from different countries contributing to the field, collaborations were largely absent. Notwithstanding the regional and contextual differences impacting their studies, previous authors have validated the positive impact of DHRM on organizations [105]. Additionally, few authors proposed new models and theories for implementing DHRM tools and techniques. Moreover, the cluster analysis (Figures 13, 14, and 15) indicates that the existing literature covers more of the theoretical aspect of DHRM than the technical and technological parts. Hence, one best-most effective model or framework depending on a certain number of technologies for all organizations globally cannot be suggested.

Current context and future research directions

The present study conducted a bibliometric analysis, a systematic literature review, and cluster analysis to discern the current research context and future directions in the field of DHRM. The bibliometric analysis revealed that the field is relatively juvenile and lacks diversity, thus presenting ample opportunities for future research. The cluster analysis (figure 13) demonstrated that the larger cluster of research sub-areas, colored red, represents the established themes in the field, while the smaller blue cluster represents potential research gaps. The established research sub-areas are primarily centered around information systems, knowledge management, employment, personnel, information management, and personal training, with a notable absence of sub-areas concerning employee satisfaction, engagement, retention, and outcomes of DHRM. Apart from these clusters, the existing definition can be separated into three different layers, namely operational, relational, and transformational levels. The nuances of the definition and their manifestation are summarized in Table 5.

The thematic analysis corroborated the bibliometric and cluster analyses, highlighting information systems as one of the most dominant themes in the field, along with human resource management and e-HRM. Additionally, the thematic mapping revealed the lack of diversity in the field, with the majority of the themes being closely associated with a limited set of keywords. Additionally, the reviewed literature presented several future research directions. These are grouped as eight categories and presented in Table 6.

Thus, the present study provides evidence that the field of DHRM requires more diversity in research topics, sub-areas, and themes to broaden the scope of the field and address the gaps in knowledge. Future research should aim to address the lack of diversity in the field, emphasizing sub-areas related to employee satisfaction, engagement, retention, and outcomes of DHRM. Such research endeavors would not only facilitate a more

Table 4 Systematic review table for DHRM

Author(s)	Title	Type of methodology (research method)	Focus area	Level and type of analysis	Type of respondents	Theoretical logic and framework	Key findings
Jani, Muduli [106].	Human resource transformation in India: examining the role digital human resource technology and human resource role	Post-positivist methodology; survey method	Impact of HR role and digital HR technology on successful HR transformation.	Organizational Level; AMOS and structural equation modeling	918 executives from Fortune 500 Indian companies.	N/A	HR transforming using digital HRM can significantly enhance business outcome of Fortune 500 companies of India. HR role (e.g., strategic, employee champion, change agent, and administrative expert) acts as a mediator. Human resource transformation requires optimization of specific roles.
Sharma, Tyagi [51]	Review on Digitalisation and Artificial Intelligence in Human Resource Function of Energy Sector	Key Informant Interview	Linkages between digital human resource management and people's performance in the Indian energy sector; the role of digitalization and artificial intelligence in building competitive advantage.	Organizational level; qualitative analysis	Experts from the energy sector	N/A	The prerequisites for a successful digitalization are continuously improving and exploring unexplored frontiers for managing HR Digitally for HR Excellence and Sustainable Business Performance
Ghazy and Fedorova [107]	Digital HRM and Hotel Business: A Global Bibliometric Analysis	Bibliometric analysis	Challenges and main trends in the hotel digital HRM	Bibliometric analysis	N/A	N/A	Current context of the field and recommendations for future studies
Khatib and Alshawabkeh [108]	Digital Transformation and Its Impact on Strategic Supremacy Mediating Role of Digital HRM: an Evidence from Palestine	Quantitative approach	Digital transformation in achieving strategic sovereignty in a world that is intensely competitive and in a digitalized era	Organizational level; smart PLS program-structural equation modeling	420 top and middle managers in two mobile telecommunications companies in Palestine	N/A	The positive and good impact of digital transformation and digital human resource management (mediating variable) on strategic supremacy;
Taule, Følstad [109]	How Can a Chatbot Support Human Resource Management? Exploring the Operational Interplay	Qualitative approach	The operational interplay between the chatbot, the HRM function, and the organization at large	Organizational level	13 HRM practitioners	N/A	The chatbot impacts the HRM function in terms of new tasks and competence requirements

Table 4 (continued)

Author(s)	Title	Type of methodology (research method)	Focus area	Level and type of analysis	Type of respondents	Theoretical logic and framework	Key findings
B. Holm [110]	Institutional context and e-recruitment practices of Danish organizations	empirical; interpretive methods	Recent changes in recruitment practices of Danish organizations and the adoption of e-recruitment from a macro-organizational perspective	Organizational level; inductive analysis of data	Exploratory interviews and observations of the associated people	Institutional theory	Recruitment practices are strongly influenced by changes stemming from functional and social pressures of the wider society; the recruitment field has transformed and reviewed its practices to a great extent due to the changes in how individuals search for employment and expect to be hired
Barykin, Kalinina [111]	Personnel management digital model based on the social profiles' analysis	Evidence-based	Examining a new approach to defining the task of personnel management in the digital age, taking into account the fundamental changes in internal communications stemming from the increasing use of digital platforms	Organizational level	N/A	N/A	Theoretical foundations for developing a model for personnel management with consideration of the distinct employee features based on social media data and included in the assessment process.
Iyer, Pani [112]	User Adoption of e-HRM—An Empirical Investigation of Individual Adoption Factors Using Technology Acceptance Model	Qualitative approach	Identifying and testing the factors influencing e-HRM adoption in terms of "intention to use" and "actual usage behavior."	Organizational level; 10 item scale	HRM professionals	TAM	Accept TAM2 hypotheses
Fisher and Cassidy [93]	Use of Relational e-HRM Tools in Gig Worker Platforms	Inductive	How gig work platforms currently use technologies for managing the relationship with gig workers.	Organizational Level	N/A	Self-Determination Theory	e-HRM systems more focused on relational functions could be used to help gig workers meet their relatedness needs
Arijomandy [113]	Social media integration in electronic human resource management: Development of a social e-HRM framework	Qualitative approach	Electronic human resource management (e-HRM) practices involving social media	Organizational level	16 organizations	Proposed Social Participation Framework	Four alternative models

Table 4 (continued)

Author(s)	Title	Type of methodology (research method)	Focus area	Level and type of analysis	Type of respondents	Theoretical logic and framework	Key findings
Simmonds and Zemit Lupi [114]	Improving the matching process of e-mentoring in luxury hotels	Survey method; quantitative and qualitative	The effectiveness of an innovative e-mentoring program introduced in an international group of luxury hotels	Quantitative and qualitative data analysis	193 people engaged in a pilot e-mentoring program		While mentoring programs have generally gained increasing popularity in various organizations, in the recent past, the success of e-mentoring has attracted little empirical research. E-mentoring is a vehicle for providing a guided mentoring relationship over large distances, largely through email, but also by using technology, including the voice over internet protocol (VOIP)
Amoako, Jiang [115]	Examining the Effect of Organizational Leadership, Organizational Structure, and Employee Technological Capability on the Success of Electronic Human Resource Management	Empirical method	The effect of organizational leadership, organizational structure, and employee technological capabilities on the implementation success of e-HRM system	Organizational level	N/A	N/A	Organizational structure and employee technological efficacy/competence have a positive and significant influence on the successful implementation of e-HRM. Conversely, organizational leadership did not significantly influence the success of e-HRM. However, it must be acknowledged that internal marketing plays a critical role in aligning organizational goals, employee capabilities, and expectation

Table 4 (continued)

Author(s)	Title	Type of methodology (research method)	Focus area	Level and type of analysis	Type of respondents	Theoretical logic and framework	Key findings
Alkhodary [116]	The impact of e-hrm on corporates sustainability: A study on the SMEs in Jordan	Qualitative approach	The impact of electronic human resources management (E-HRM) on corporates' sustainability (CS)	Organizational level	500 respondents from small medium enterprises SMEs in Jordan, 385 responses were valid.	N/A	There is an impact of E-HRM on corporates' sustainability on the SMEs in Jordan; the positive impact of E-recruitment, E-performance management, and E-learning on CS in the SMEs in Jordan

Table 5 Three-layer definition framework for DHRM

Perspective	Definition and scope	Typical digital tools/activities	Key value/benefits	Representative studies
Operational	Digitization and automation of core, and transactional HR processes, routine, repetitive tasks.	E-recruitment, e-selection, payroll systems, chatbots for FAQs, time and attendance management	Increased efficiency, accuracy, cost savings, and reduced administrative burden for HR staff.	Sharma, Tyagi [51]; Al-Hyari [117], Kuchеров and Tsybova [118], Islam, and Rahman [119]
Relational	Use of digital platforms to enhance communication, networking, and relationship management among HR, employees, and managers.	HR portals, internal social media, e-learning platforms, e-mentoring, and relational e-HRM for gig platforms	Enhanced engagement, improved knowledge sharing, better relationship management, and stronger employer branding.	Fisher and Cassidy [93], Arjonandy [120], Wiblen [121], and B. Holm [110]
Transformational	Strategic use of digital technologies to fundamentally change HR's role, supporting business transformation and enabling evidence-based, data-driven HR strategies and analytics.	HR analytics, AI-driven talent management, cloud-based HRM suites, mobile HR apps, and digital HR strategy	Data-driven decision-making, strategic alignment of HR with business, organizational agility, and support for innovation and change.	Jani, Muduli [31], Wang [122], Zavyalova, Sokolov [123], Ghazy and Fedorova [107]

Table 6 Future research directions

Thematic group	Future research directions	Sources
1. Cross-contextual and sectoral analysis	1. Extend research to international and cross-sector contexts to understand the impact and adoption of digital HRM in various industries and countries. 2. Validate models and frameworks in different sectors and organizational types.	Schmidt, Pohler [69], Bondarouk, Schilling [124], Manivannan, and Valiammal [125]
2. Technology Implementation and Adoption Challenges	1. Explore barriers, enablers, and best practices for E-HRM, RPA, and AI adoption before/during implementation. 2. Study user satisfaction, acceptance, and resistance across contexts. 3. Investigate critical factors for successful digital transformation in HR.	Jani, Muduli [31], Agarwal, Kapoor [126], Arora, Prakash [127], and Sharma and Sharma [128]
3. Strategic and organizational Impact	1. Investigate the role of HRM capability maturity, leadership, structure, and culture on digital HRM effectiveness. 2. Analyze mediation effects of HR roles in digital transformation. 3. Examine impact on organizational performance, sustainability, and strategic advantage.	Jani, Muduli [31], Amoako, Jiang [115], and Wang [122]
4. Talent management, analytics, and gamification	1. Research the effectiveness of HR analytics, gamified employer branding, and advanced data-driven approaches in talent acquisition and management. 2. Study the human-technology interface in talent management. 3. Develop new models for talent identification and analytics use.	Borisova, Silayeva [129], Wiblen and Marler [130], Shree and Singh [131], and Silic, Marzi [132]
5. Social media, communication and employee behavior	1. Examine both positive and negative (dark side) impacts of social networking and digital communication tools in E-HRM, including workplace deviance and relatedness. 2. Study integration of social media with e-HRM and its organizational implications.	Fisher and Cassidy [93], Arjomandy [120], and Nayak, Budhwar [133]
6. Remote/agile working and digital roles	1. Investigate the evolution of remote/agile working practices post-pandemic. 2. Identify new digital HR roles, tasks, and competencies for supporting agile organizations. 3. Study the effectiveness of remote work policies and e-HRM support for distributed teams.	D'Auria, De Chiara [16], Hussein and Jaaffar [134], Gouda and Tiwari [135], and McIver, Lengnick-Hall [136]
7. Ethics, privacy, and data security	1. Further explore ethical, privacy, and security challenges in digital recruitment and digital footprint analysis. 2. Develop guidelines for transparent and responsible digital HRM practices.	Agarwal, Kapoor [126], Nayak, Budhwar [133], Ardabili, Amin [137], Chatterjee, Chaudhuri [138], and Pillai and Srivastava [139]
8. Employee experience and user-centric design	1. Explore determinants of technology acceptance, user experience, and satisfaction with digital HRM platforms. 2. Examine the operational interplay of chatbots and other digital support tools in HR. 3. Study the role of organizational support in end-user satisfaction.	Taule, Følstad [109], Iyer, Pani [112], Manistitya and Fongsuwan [140], and Voermans and Van Veldhoven [141]

comprehensive understanding of the field, but also provide insights that could be applied in practice to enhance the effectiveness of DHRM in organizations.

Conclusion and limitations

The analyses conducted for this research suggest a steady growth in the DHRM scholarly field. From the application or implementation perspective, the benefits of the field's development are two-dimensional. First, using the knowledge and techniques derived from this development directly helps businesses make more informed decisions concerning their human capital. Second, the increased productivity and efficiency resulting from the usage directly benefit the customers in terms of product or service availability and affordability. Furthermore, the implementations facilitate the managerial process by delineating how the employee strengths and weaknesses interact against the external environments. This research contributes to the field of DHRM by revealing the existing and future research directions. Additionally, the acknowledgment of most contributing and relevant authors and sources in this field embraces the possibility of leading future contributors to amalgamate theoretical or conceptual frameworks on different aspects of the field that are already accepted in the majority portion of scientific productions. The analyses of these scientific productions published in almost 27 years show the increasing research interest and the breadth of existing knowledge. Although the field's growth manifested through the analysis, a wide array of theoretical lacking prevails, necessitating future contributions and developments.

This research identified the most cited sources and authors, dominating and emerging keywords and themes, and publication trends. As identified by the clusters (Figure 13), DHRM's focus noticeably differs from that of mainstream HRM. In addition, the systematic review (Table 4) presents a set of keywords, applied methods, findings, and researched areas that can be attributed to the current context of the field. Furthermore, this research revealed promising future research directions already outlined by existing compositions. Through cluster and theme analyses, this research furnishes the ground for future researchers to contribute through their scientific works.

However, despite the objectivist philosophy adopted by the researchers, the findings cannot omit certain limitations. First, the inclusion of all the aspects of the field cannot be ensured, as the bibliometric data and abstract of the documents were extracted only from two database (i.e., the Scopus and WOS). Additionally, the searching and filtering techniques applied while conducting this extraction may lead to the exclusion of some critical documents. The dependency and

interconnectedness of DHRM with other technological- and management science-related fields perplex the identification process of the relevant literature. Finally, the cut-off date for bibliometric data means that the most recent or in-press research may not be fully captured. Thus, the researchers recommend that future research should exploit a broader range of databases beyond Scopus and Web of Science, including regional and subject-specific repositories, to ensure a more comprehensive and inclusive collection of relevant literature. More rigorous and transparent searching and filtering techniques are also encouraged to minimize potential selection bias and capture studies employing diverse methodologies and terminologies. Furthermore, future researchers should consider the inclusion of non-English publications and gray literature to enrich the global and practical perspectives within the DHRM domain. In addition, a rigorous qualitative content analysis should be conducted on current scientific productions to identify not only the governing theories, research processes, and methods, but also to uncover the underlying assumptions, contextual factors, and conceptual linkages that may not be evident through bibliometric analysis alone. Collaboration among multidisciplinary research teams, as well as the use of advanced text-mining or natural language processing tools, could further enhance the reliability and depth of such analyses. Finally, longitudinal studies tracking the evolution of key concepts and the impact of digitalization on HRM practices over time would be valuable, as would comparative studies across different regions, industries, and organizational contexts to uncover nuanced patterns and best practices in DHRM research and implementation.

Author contributions

M.N.S. and M.S.U. contributed to conceptualization, formal analysis, and methodology; M.M.R. contributed to investigation, formal analysis, and writing-original draft; M.S.U. contributed to validation and supervision. M.N.S. and M.M.R. contributed to software, writing-review & editing, and validation.

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Declarations

Ethics approval and consent to participate

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Competing interests

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